

Meridian — Binary Stock Outcome Markets

Pre-Search Checklist (Completed)

1. Domain Selection

Domain: Finance + blockchain infrastructure + prediction markets.

Use Cases: Buy/Sell Yes contracts Buy/Sell No contracts Mint Yes/No pairs Market making Oracle settlement Redemption flows **Verification Requirements:** Yes + No payout invariant always equals \$1.00 Vault collateralization checks Immutable settlement state Oracle freshness/confidence validation **Data Sources:** Pyth Network oracle feeds Solana RPC infrastructure Order book state Wallet balances

2. Scale & Performance

Expected Scale: 100–1,000 DAU initially 10–50 concurrent traders 1–5 tx/sec MVP target **Latency Requirements:** UI updates under 250ms Order execution perceived under 1 second Settlement completed within 10 minutes after market close **Cost Constraints:** LLMs are not core infrastructure for this project. Focus budget on RPC, blockchain infra, and testing.

3. Reliability Requirements

Cost of Failure: Incorrect settlement or invariant violations could cause loss of funds and protocol instability.

Non-Negotiable Verification: Invariant testing Oracle validation Atomic mint/redeem logic Position consistency checks **Human-in-the-loop:** Admin override settlement and pause/unpause controls required.

4. Team & Skill Constraints

Required Skills: Solana + Anchor smart contracts Order book mechanics Oracle integrations Frontend wallet integrations Financial systems testing **Testing Comfort:** Strong understanding of invariant testing and integration testing required.

5. Architecture Discovery

Recommendation: Avoid agent frameworks for core trading logic. Use deterministic services and typed APIs.

Architecture: Solana smart contracts using Anchor Phoenix on-chain order book Node.js automation service Next.js frontend **State Management:** Market state, settlement state, vault balances, and oracle state must be deterministic.

6. LLM Selection

Use Case**Recommended Tool** Protocol Architecture Claude Code Frontend Development Cursor Debugging & Reviews ChatGPT Security Review Claude Code

Recommendation: Use Claude Code for smart contract correctness and Cursor for implementation speed.

7. Tool Design

Core Tools: Anchor framework Solana CLI Pyth SDK Phoenix SDK Wallet adapters **Error Handling:** Handle stale oracle data, RPC failures, settlement delays, and wallet rejections.

8. Observability Strategy

Monitoring Stack: Grafana Prometheus Sentry **Critical Metrics:** Settlement failures Invariant violations Vault mismatches Oracle failures

9. Eval Approach

Correctness Measurement: Primary metric is invariant preservation.

Testing: Automated integration testing Property-based testing Lifecycle simulation CI pipeline validation

10. Verification Design

Claims to Verify: Oracle prices are valid Markets are expired before settlement Vault balances remain collateralized Redemption values are correct **Escalation Triggers:** Oracle unavailable, stale data, settlement retries exhausted, abnormal balances.

11. Failure Mode Analysis

Failure Handling: Retry queues Admin override settlement Emergency pause Fallback RPC providers **Graceful Degradation:** Frontend should remain viewable even if trading is paused.

12. Security Considerations

Key Risks: Private key leakage Settlement manipulation Double redemption exploits Oracle spoofing **Security**

Controls: Environment-based secrets Audit logs Role-based admin permissions

13. Testing Strategy

Required Tests: Unit tests for mint/redeem/settle Integration lifecycle tests Invariant tests Adversarial attack simulations **Regression Testing:** Run all tests on every PR using GitHub Actions.

14. Open Source Planning

Recommended Open Source Components: Frontend Smart contracts SDKs Deployment scripts **License:** MIT or Apache 2.0 recommended.

15. Deployment & Operations

Recommended Infrastructure: Frontend: Vercel Automation Service: Railway or Fly.io RPC: Helius **Operations:** CI/CD pipelines, monitoring dashboards, rollback procedures, and upgrade controls required.

16. Iteration Planning

Feedback Collection: Discord community Testnet user analytics Transaction failure analysis **Roadmap Priorities:** Correctness Settlement reliability Trading UX Liquidity and analytics